CS472 - Computer Networks: Theory, Applications and Programming
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Syllabus

Couse Objectives
Introduction to computer networking theory, applications, and programming, focusing on large heterogeneous networks. Broad topdown introductions to computer networking concepts including distributed applications, socket programming, operating systems, router support, router algorithms, and sending bits over congested, noisy and unreliable communication links.

This is a hands-on course. The lectures will have tooling and programming demonstration and you will be expected to do similar things for homework and project assignments.

Student Expectations / Prerequisites
This is a hands-on computer science systems class. As such student must be comfortable working with a systems programming language such as C for the programming component of this course. The student should also have access and plan to use a Linux environment to complete required coursework. Students may choose the Linux environment that they want to use for development. Acceptable options are Tux – CCI's Linux cluster, or a local VM on your personal computer. Please note that if you choose to run your own virtual machine you will be responsible for supporting it. Some good suggestions for free local virtual machine solutions include Orbstack for MacOS - this is what I use. Note that Orbstack is commercial software but they provide a free license for personal, non-commercial work. Windows users may use the Linux Subsystem for Windows.

Office Hours
I conduct office hours both in person and virtual. My office hours for the term, both in-person and virtual over zoom will be posted on Blackboard under the Instructor Information tab.

Discord
I do not like using emails to manage classes, its not designed for that. Blackboard announcements are also a drag. As such, we have a class Discord that is my preferred venue for communicating with the class. You can DM me over Discord or better yet, use the channels that are open to the entire class to ask questions. If you have a question, share with the class, somebody else also likely has that question.

Topics
1. Introduction – OSI Reference Model, TCP/IP Intro, RFCs, Socket Review
2. Network Definition (and evolution)
3. Network Concepts (packet and network switching, latency, resiliency, packets, encapsulation, etc)
4. Network Standards (e.g., RFCs)

5. Network Protocols and PDUs
6. OSI Conceptual Reference Architecture
7. Concreate IP Network Architecture: Ethernet, TCP/IP, UDP/IP, IPv4, IPv6
8. IP Addressing (IP Address Classes, CIDR, ports, NAT, etc)
9. Socket Programming (Systems Programming)
10. Higher Level Network Programming (HTTP Frameworks, GRPC)
11. Application Layer – HTTP, FTP, DHCP, DNS
12. Transport Layer – TCP & UDP – Deep Dive into TCP
13. Network Layer – deep dive into IP
14. Link Layer (frames, error detection, ethernet)
15. Network Security - TLS
16. Routing
17. Modern Network Considerations (WiFi and Mobile)
18. Software Defined Networks (including Cloud Networking)

Expected Learning
By the end of this course the student should be expected to have a solid grasp on:

- Various Networking Protocols, and how these protocols work together
- The OSI and TCP/IP Reference Model
- Key network addressing concepts, IP addresses, CIDR, MAC addresses
- The difference between connectionless and connection-oriented protocols
- How protocols are designed and support each other, both horizontally and vertically.
- Modern network architecture concerns – security, reliability, resiliency, congestion control.
- How data gets routed around networks and different types of routing protocols
- Key aspects of virtual and physical network hardware – routers, switches, WiFi, etc
- Hands on building of several different types of network protocols via the socket API.

Textbook

Starting in 2024, I will no longer be requiring a course textbook. That said, a highly recommended one that I will reserve that will complement learners that benefit from having a textbook is below.

Computer Networking, A Top Down Approach, Kurose & Ross, 8th Edition.
Note that if you can find the 7th edition cheaper you should be fine.

We will be starting off with a broad overview of computer networks and then traverse topics starting at the high level (network concerns closest to the application), then work down the stack to the data link layer covering how bits flow over the network.

The free online resources listed below will also be useful or helpful in this course – they are also linked under the "Useful Resources" tab on blackboard:

1. Computer Networks: A Systems Approach - https://book.systemsapproach.org/
2. Computer Networks, Principals, Protocols & Practice - https://github.com/cnp3/ebook/releases/download/draft-3rd/CNP3-2021.pdf
3. Beej's Guide to Network Programming Using Internet Sockets - https://beej.us/guide/bgnet/

Tools (get these installed in your OS)

1. "Unix" command line tools, e.g., netstat, arp, traceroute, ifconfig, etc
2. Wireshark
3. C Programming environment (I use VSCode with gcc on my machine)
4. During the end of the term we will be using GNS3 for a course routing lab. You do not need to install this on your computer, CCI provided virtual machines will be used.

Programming Expectations

This full title of this course is Computer Networks: Theory, Applications, and Programming. As such, this course will require you to use a systems programming language to get practical hands-on experience working with network protocols. I will be using C in this course and will accept homework in either C or C++. Many of the assignments will prove you with scaffolded code to get you started, my code will be provided in 'C'. We will be reviewing C programming concepts in this class around socket programming and buffer manipulation in case you need to dust off your C skills.

Throughout this course I will also be demonstrating some newer systems programming languages that have strong networking support such as Go and Rust.

Assignments, Programming Assignments and Labs.

This course will mainly consist of programming assignments that have both a programming and non-programming component. Other assignments will be assigned to promote understanding of foundational concepts, and gaining hands on experience in a networking lab where you will be configuring a virtual software router.

Assignments, Due Date and your Late Day Bank

This course will have assignments that will be spread out over the course as mentioned above. Given my previous experience, students tend to have different levels of readiness, especially with background in programming. I will adjust deadlines for the whole class as necessary, based on my observation of student effort and questions being asked over Discord, and attending office hours (mine or the TAs). One off deadline extensions are not possible, but I do realize that things come up from time to time. If you go into grade center, you will notice that you have a bank of 5 late days to use over the term. Late submissions will be accepted without penalty by deducting from your late day bank. Once you exhaust your bank, late assignments will take a 50% penalty on day 1, and will not be accepted on day 2+. Assignments are typically due at the end of the week (midnight on Friday).

Note that the late day bank is to accommodate students “when things come up” for whatever reason. This includes work deadlines, needs of other courses, sickness, unexpected travel, and so on. They should be considered as “insurance”, because once they are gone, they are gone. You do not require any permission to submit assignments late – they will be accepted without penalty, without any reason for lateness so long as late days exist in your bank.

Expectations of Timely Communication for Hardships
If, for whatever reason, you find yourself struggling with keeping up with course deliverables, it is important that you let me know right away, including your plan to get back on track. I am here to help you and am willing to be as flexible as possible. However, my flexibility becomes very limited if you come to me weeks after one or more assignment deadlines have been missed with a hardship story expecting me to accept missing work without significant penalty, if even at all.

Grading Breakdown
70% homework, programming assignments, and labs
30% Quizzes (a total of 5)

Note that there will be 5 online quizzes over the course of the term. Each quiz will be worth 25 points. Your final quiz score will be calculated as follows. Top 4 quiz scores + (0.5) lowest quiz score. In other words, your quiz score will be based on your top 4 quiz scores, rather than dropping the lowest, I will be providing 50% of your worst quiz score as extra credit. Thus your maximum total quiz score will be 112.5 / 100. If you are satisfied with your first 4 quiz scores, you can skip the last one. This will then be weighted 40% of your total course score.

Rough Schedule
The following is a rough schedule for this course, by week:

Introduction to Computer Networks, Socket Programming Tutorial
1. Introduction to Network Protocols
2. Network Programming with Sockets (UDP & TCP)
3. Internet and Networking Concepts
4. Application Layer Part 1 - Concepts
5. Application Layer Part 2 – Deep Dive into the HTTP Protocol Suite
6. Transport Layer Concepts
7. Transport Layer Deep Dive into TCP and UDP
8. Network Layer - IP Concepts
9. Network Layer - Routing Protocols
10. In class Networking Lab

Grades
A+ (98-100); A (93-97); A- (90-92)
B+ (87-89); B (83-86); B- (80-82)
C+ (77-79); C (73-76); C- (70-72)
D (60-69)
F (< 60)

University Policies:
This course follows university, college, and department policies, including but not limited to:

- Academic Integrity, Plagiarism, Dishonesty and Cheating Policy: http://www.drexel.edu/provost/policies/academic_dishonesty.asp
- Student Life Honesty Policy from Judicial Affairs: http://www.drexel.edu/provost/policies/academic-integrity
- Students with Disability Statement: http://drexel.edu/oed/disabilityResources/students/
- Course Add/Drop Policy: http://www.drexel.edu/provost/policies/course-add-drop
- Course Withdrawal Policy: http://drexel.edu/provost/policies/course-withdrawal
- Department Academic Integrity Policy: http://drexel.edu/cci/resources/current-students/undergraduate/policies/cs-academic-integrity/
- Drexel Student Learning Priorities: http://drexel.edu/provost/assessment/outcomes/dslp/
- Office of Disability Resources: http://www.drexel.edu/ods/student_reg.html

Students requesting accommodations due to a disability at Drexel University need to request a current Accommodations Verification Letter (AVL) in the ClockWork database before accommodations can be made. These requests are received by Disability Resources (DR), who then issues the AVL to the appropriate contacts. For additional information, visit the DR website at drexel.edu/oed/disabilityResources/overview/, or contact DR for more information by phone at 215.895.1401, or by email at disability@drexel.edu.

Couse Change Policy
The instructor may, at their discretion, change any part of the course during the term, including assignments, grade brakdowns, due-dates, and the schedule. Such changes will be communicated to students via the course web site Announcements page. This page should be checked regularly and frequently for such changes and announcements. Other announcements, although rare, may include class cancellations and other urgent announcements will be communicated via the course discord channel.